

Note 2.5 Continuity

Continuity at a Point

- **Definition:**

- If $\lim_{x \rightarrow c} f(x) = f(c)$, then we say $f(x)$ is continuous at $x = c$ and $x = c$ is a continuous point.

$$\lim_{\Delta x \rightarrow 0} (f(x_0 + \Delta x) - f(x_0)) = 0$$

- $$\Leftrightarrow \lim_{\Delta x \rightarrow 0} f(x_0 + \Delta x) = f(x_0)$$
$$\Leftrightarrow \lim_{h \rightarrow 0} f(h) = f(x_0) (h = x_0 + \Delta x)$$

- If $\lim_{x \rightarrow c^+} f(x) = f(c)$ (or $\lim_{x \rightarrow c^-} f(x) = f(c)$), we say $f(x)$ is right (or left) continuous at c .

- **Continuity Test:**

- $f(x)$ is defined at c .
- $\lim_{x \rightarrow c} f(x)$ exists.
- $\lim_{x \rightarrow c} f(x) = f(c)$.

- **Categories of discontinuity: (c is a discontinuous point in the following)**

- removable discontinuity: $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) \neq f(c)$
- jump discontinuity: $\lim_{x \rightarrow c^+} f(x) \neq \lim_{x \rightarrow c^-} f(x)$
- infinite discontinuity: $\lim_{x \rightarrow c^+} f(x) = \infty$ or $\lim_{x \rightarrow c^-} f(x) = \infty$
- oscillating discontinuity: It oscillates too much to have a limit as $x \rightarrow c$. ($f(x) = \sin \frac{1}{x}$)

Continuous Functions

- **Continuous on an interval:**

$f(x)$ is continuous on $[a, b] \Leftrightarrow f(x)$ is continuous at every point $c \in [a, b]$, and left continuous at $x = b$, right continuous at $x = a$.

- **Continuous function:**

A continuous function is continuous at every point of its domain.

- **Theorem--Properties of Continuous Functions**

If the functions f and g are continuous at $x = c$, then the same combinations as [limit laws](#) are continuous at $x = c$.

- **Thomae's Function:**

$$f(x) = \begin{cases} \frac{1}{q}, x = \frac{p}{q} (p, q \in \mathbb{Z}, q > 0, (p, q) = 1) \\ 0, x \text{ is irrational.} \end{cases}$$

the function is discontinuous at x which is rational and continuous at x which is irrational.

- For irrational number, suppose that c is irrational, then for any possible ϵ , there will be finite positive integers q such that $\frac{1}{q} \geq \epsilon$. (that is $q \leq \frac{1}{\epsilon}$) Thus in the neighborhood of c , there exists finite rational number whose denominator $\leq \frac{1}{\epsilon}$. Then take a small enough δ such that $(c - \delta, c + \delta)$ doesn't contain those finite rational number. So that is for any $\epsilon > 0$, there exists a corresponding δ such that $0 < |x - c| < \delta \implies |f(x) - L| < \epsilon$. So $\lim_{x \rightarrow c} f(x) = f(c) = 0$ and the function is continuous as x is irrational.
- For rational number, we can take c as rational number. We can take $\epsilon = \frac{1}{2q}$, then for any neighborhood $(c - \delta, c + \delta)$, there exists a irrational number a such that $|f(a) - f(c)| = \frac{1}{q} > \epsilon$. So $f(x)$ is discontinuous as x is rational.

Inverse Functions and Continuity

- The inverse function of any function continuous on an interval is continuous over its domain.

Composites

- **Theorem--Composite of Continuous Functions:**

If f is continuous at c and g is continuous at $f(c)$, then the composite of $g \circ f$ is continuous at c .

Note: Elementary functions **on the interval contained** in its domain are continuous. (e.g: $f(x) = \sqrt{\cos x - 1}$, $D(f) = \{x | x = 2k\pi, k \in \mathbb{Z}\}$ is discontinuous)

- **Theorem--Limit of Continuous Functions:**

If g is continuous at the point b and $\lim_{x \rightarrow c} f(x) = b$, then

$$\lim_{x \rightarrow c} g(f(x)) = g(b) = g(\lim_{x \rightarrow c} f(x))$$

- **Theorem**

If $f(x) \in C([a, b])$, ($C([a, b])$ is the set of all continuous functions on $[a, b]$) then

- $f(x)$ is bounded on $[a, b]$, that is, there exists a $C > 0$ such that $|f(x)| \leq C$ for all $x \in [a, b]$.
- $f(x)$ can get its maximum value and minimum value on $[a, b]$, that is, there exists $x_1, x_2 \in [a, b]$ such that

$$f(x_1) = \max\{f(x) | x \in [a, b]\} \triangleq M, f(x_2) = \min\{f(x) | x \in [a, b]\} \triangleq m$$

(the symbol $\triangleq$ represents "defined as")

Intermediate Value Theorem for Continuous Functions

- **Definition:**

If f is continuous on a interval $[a, b]$, then for each N between $f(a)$ and $f(b)$, there exists one or more $x_0 \in [a, b]$ such that $f(x_0) = N$.

- **Corollary:**

If $f(x) \in C([a, b])$, $f(a)f(b) < 0$, then there exists $x_0 \in [a, b]$ such that $f(x_0) = 0$. This corollary to ensure the existence of zero point in $[a, b]$.